Bayesian inference of exponential random graph models for large social networks

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Abstract

This paper addresses the issue of sampling from the posterior distribution of exponential random graph (ERG) models and other statistical models with intractable normalizing constants. Existing methods based on exact sampling are either infeasible or require very long computing time. We propose and study a general framework of approximate MCMC sampling from these posterior distributions. We also develop a new Metropolis-Hastings kernel with improved mixing properties, to sample sparse large networks from ERG models. We illustrate the proposed methods on several examples. In particular, we combine both algorithms to fit the Faux Magnolia high school data set of Goodreau et al. (2008), a network data with $1,461$ nodes.

Keywords: Exponential random graph model, Bayesian inference, Intractable normalizing constant, Markov chain Monte Carlo

1. Introduction

Networks are used to represent relations or connections among various types of entities, e.g., interconnected web sites, airline networks, and electricity networks. Network analysis has a broad range of applications including social science, epidemics, language processing, and etc. This paper focuses on social networks where the nodes of the network typically represent individuals or other social entities and the edges capture relationships such as friendship and collaboration.

A network x with n nodes is a $n \times n$ matrix, where entry x_{ij} represents the strength of the connection between the ordered pair of nodes (i, j), and can take a finite number of values. Let $\mathcal{X}$ be the space of all such networks (we omit the dependence on n). In the simplest case where the network is undirected and binary, $x_{ij} = x_{ji} = 1$ if there is an edge between nodes i and j (in

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other words, i and j are neighbors), and 0 otherwise. In this case, $\mathcal{X}$ is the space of all triangular matrices $\{x_{ij},\ 1 \leq i < j \leq n\}$ with 0-1 entries. The specific modeling framework for social networks considered in this paper is the exponential random graph (ERG) model (Wasserman and Pattison, 1996). It assumes a probability distribution for x given by

$$p_\theta(x) = \frac{f_\theta(x)}{Z(\theta)} = \frac{1}{Z(\theta)} \exp\left\{\sum_{k=1}^{K} \theta_k S_k(x)\right\}, \quad x \in \mathcal{X}, \tag{1}$$

where θ represents the unknown parameters, $f_\theta(x)$ is the un-normalized likelihood, and $Z(\theta)$ is the normalizing constant,

$$Z(\theta) = \sum_{x\in\mathcal{X}} \exp\left\{\sum_{i=k}^{K} \theta_k S_k(x)\right\}. \tag{2}$$

$S_k(x)$ is a network statistic of interest, for example, the number of edges $E(x) = \sum_{i,j} x_{ij}$ to capture network density; the number of triangles $\sum_{i,j,h} x_{ij}x_{jh}x_{hi}$ to capture transitivity; the number of 2-stars $\sum_{i,j,h} x_{ih}x_{jh}$, where a k-star ($k \geq 2$) is a node with k neighbors or a node of degree k. See Wasserman and Pattison (1996); Snijders et al. (2006); Hunter and Handcock (2006) for more examples.

This paper proposes a flexible and efficient algorithm to sample from the posterior distribution of the parameter θ in ERG models. The algorithm also applies to other statistical models where the intractable normalizing constant is an issue. We also propose a new Metropolis-Hastings (M-H) sampler to sample efficiently from the ERG distribution p_θ on the space of networks $\mathcal{X}$. The new algorithm is particularly effective in dealing with large sparse social networks.

1.1. MCMC for models with intractable normalizing constants

One of the main issue with ERG models is the intractability of the normalizing constant $Z(\theta)$. Evaluating $Z(\theta)$ is simply infeasible for most networks in practice. For an undirected binary network of 10 people, the calculation of $Z(\theta)$ may involve 2^{45} terms. This problem, which becomes more severe for large networks, implies that the likelihood function and the posterior distribution (in a Bayesian framework) cannot be evaluated, even up to a normalizing constant.

Because of the intractable function $Z(\theta)$, computing the maximum likelihood estimator (MLE) for this model is not straightforward. A solution dating back to Besag (1974) is pseudo-likelihood methods where the likelihood function is replaced by a more tractable approximation (Strauss and Ikeda, 1990). But this often does not work well in practice. More recently, algorithms based on stochastic approximation (Younes, 1988) and MCMC (Geyer and Thompson, 1992; Hunter and Handcock, 2006) have been developed that make it possible to compute the MLE reliably. However, the behavior (particularly the asymptotic behavior) of the MLE for this type of statistical models is still poorly understood. This fact makes the Bayesian inference for ERG models particularly attractive.

Here we take a Bayesian approach. We assume that the parameter space Θ is a subset of the p-dimensional Euclidean space $\mathbb{R}^p$. Let us denote the observed network by $\mathcal{D}$, and μ the density (wrt the Lebesgue measure on Θ) of the prior distribution for θ on the parameter space Θ. Then the posterior distribution has density

$$\pi(\theta|\mathcal{D}) = \frac{1}{\pi(\mathcal{D})}\frac{1}{Z(\theta)}f_\theta(\mathcal{D})\mu(\theta), \quad \theta \in \Theta,$$

where $\pi(\mathcal{D})$ is the normalizing constant of the posterior distribution (or the marginal distribution of $\mathcal{D}$). There are two intractable normalizing constants in the posterior $\pi(\theta|\mathcal{D})$, and Murray et al. (2006) coined the term doubly-intractable distribution to refer to this type of distributions. Conventional MCMC methods can get rid of $\pi(\mathcal{D})$ but not $Z(\theta)$. For example, in the Metropolis-Hastings algorithm, given the current θ and a new value θ' proposed from $Q(\theta, \theta')$, the acceptance ratio is

$$\min\left[1, \frac{\pi(\mathcal{D})}{\pi(\mathcal{D})}\frac{f_{\theta'}(\mathcal{D})}{f_\theta(\mathcal{D})}\frac{Z(\theta)}{Z(\theta')}\frac{\mu(\theta')}{\mu(\theta)}\frac{Q(\theta',\theta)}{Q(\theta,\theta')}\right] = \min\left[1, \frac{f_{\theta'}(\mathcal{D})}{f_\theta(\mathcal{D})}\frac{Z(\theta)}{Z(\theta')}\frac{\mu(\theta')}{\mu(\theta)}\frac{Q(\theta',\theta)}{Q(\theta,\theta')}\right]. \quad (3)$$

While $\pi(\mathcal{D})$ disappears, $Z(\theta)$ remains in the Hastings ratio.

An early attempt to deal with this issue is to estimate $Z(\theta)$ beforehand with methods such as bridge sampling (Meng and Wong, 1996) or path sampling (Gelman and Meng, 1998), and then plug it into the acceptance ratio (e.g., Green and Richardson, 2002). However, the Markov chain may not converge to the target distribution because of the estimation error, and computation is unnecessarily intensive. Another way is to replace the likelihood with pseudolikelihood, and carry out inference on the induced pseudo-posterior. Yet the pseudo-posterior is not the posterior of interest, and it may not be a proper distribution.

Asymptotically correct MCMC methods were developed recently. Møller et al. (2006) ingeniously used an Auxiliary Variable Method (AVM) to replace the intractable normalizing problem with an exact sampling (or perfect sampling, Propp and Wilson 1996) problem from p_θ. Murray et al. (2006) proposed exchange algorithms to further improve its mixing. However, exact sampling is quite computationally expensive and is simply infeasible for many useful models (e.g., the ERG model (17) due to alternating k-triangle statistic (15)). Atchadé et al. (2008) developed an adaptive MCMC method which estimates $Z(\theta)$ within the MCMC procedure. Their algorithm does not require perfect sampling but remains computer-intensive, especially for large networks.

In this paper, we propose a general framework of approximate MCMC algorithms for doubly-intractable distributions. The idea is built around the AVM of Møller et al. (2006) and the exchange algorithm of Murray et al. (2006). But we avoid the exact sampling from p_θ which is replaced by samples from a Markov kernel that is close to p_θ. The resulting algorithm is significantly faster and we prove that it is stochastically stable in the sense that it admits an invariant distribution. Furthermore the distance between the invariant distribution of the algorithm and the target posterior distribution can be controlled by the user and be arbitrary small (of course, at the expense of more computing time).

The idea formalizes a practice that is already common among practitioners dealing with doubly-intractable distributions. For example, it has been advocated recently by Jin and Liang (2009); Liang (2010); Caimo and Friel (2011). In this paper, we carefully analyze these approximate schemes and present some theoretical results to guide their use in practice.

1.2. Simulating large social networks

An important tool in computing with ERG models is the ability to sample from p_θ, the ERG model itself for a given parameter value. This applies to both frequentist and Bayesian approaches. Various Metropolis-Hastings (MH) algorithms are often used for the task. The mixing of these algorithms depends upon the choice of proposals or transition kernels used to generate a new network. See Snijders (2002) for a detailed discussion and a variety of proposals. For small to moderate networks, existing proposals usually work well. However, for large and sparse networks, without proper adjustment of the proposals, these algorithms may suffer from slow mixing.

One problem documented in Morris et al. (2008) has to do with the widely used random sampler proposal for simulating ERG models. In random sampler, one randomly selects a dyad (i.e. a pair of nodes) to toggle (from edge to no edge, or vice versa). Sparse networks have much more disconnected dyads than connected ones, thus oftentimes the proposal will add an edge to the network. If the parameter corresponding to network density is negative (which is usually the case for real-world large networks), then most of the time the proposed networks will be rejected, and the Markov chain will stay on the same state for a long time. A refined proposal called TNT (tie-no tie, Morris et al. 2008) improves the mixing by holding a probability 0.5 of choosing an edge and proposing to delete it. The rest of the time it picks an empty dyad to toggle.

Another problem resides in models with higher order network statistics (as opposed to nodal or dyadic statistics), for instance the number of triangles and alternating k-triangles. In the simulation of sparse large networks, proposals such as random sampler or TNT cannot vary the values of these statistics efficiently. To construct a triangle from three disconnected nodes, all three pairwise dyads need to be chosen, which takes many steps to achieve. If the statistic value rarely changes, the corresponding parameter has limited effect on the simulation. It is not hard to imagine that the time to reach convergence of the ERG model will be quite long.

In this paper, we are particularly interested in the mixing of ERG models involving triangles. Triangles are usually used to characterize transitivity, which is generally of interest in social network modeling. Recently, model degeneracy (Handcock, 2003; Snijders et al., 2006) has caught researchers' attention, where the model places disproportionate probability on a small set of outcomes. An important statistic to help push back model degeneracy, alternating k-triangle (Snijders et al., 2006), also consists of triangles. Here we introduce a new M-H proposal called OTNT (open triangle-tie-no tie) to sample more efficiently from p_θ for large sparse social networks.

The remaining of the paper is organized as follows. In section 2, we propose a general framework for approximate MCMC algorithms and explore the theoretical properties. In section 3, we describe the M-H move OTNT. A simulation study and real data analysis are presented in section 4. We give detailed proofs in the appendix.

2. Approximate MCMC simulation from doubly-intractable distributions

Let $\{p_\theta(\cdot),\ \theta \in \Theta\}$ be a statistical model with sample space $\mathcal{X}$ and parameter space Θ. As in the exponential random graph model, we assume that $\mathcal{X}$ is a finite set. But for notational convenience, we will still write $\int_{\mathcal{X}} f(x)dx$ to denote the summation $\sum_{x\in\mathcal{X}} f(x)$. We also assume that for all $\theta \in \Theta$, $p_\theta(\cdot) = Z(\theta)^{-1} f_\theta(\cdot)$, for some normalizing constant $Z(\theta)$ which is intractable. The parameter space Θ is a subset of the p-dimensional Euclidean space $\mathbb{R}^p$ endowed with the Lebesgue measure. Suppose that we observe a data set $\mathcal{D} \in \mathcal{X}$ and choose a prior density μ (wrt the Lebesgue measure on Θ). The posterior distribution is $\pi(\theta|\mathcal{D}) \propto f_\theta(\mathcal{D})\mu(\theta)/Z(\theta)$. As $Z(\theta)$ cannot be evaluated, direct simulation from $\pi(\theta|\mathcal{D})$ is infeasible. The goal is to propose Monte Carlo methods that are consistent and perform well in applications.

As mentioned in section 1.1, the acceptance probability (3) of a straightforward M-H to sample from $\pi(\theta|\mathcal{D})$ is intractable as it involves the intractable normalizing constants. Møller et al. (2006) circumvent the problem by introducing a joint distribution $\tilde{\pi}(\theta, x|\mathcal{D}) = \pi(\theta|\mathcal{D})\tilde{f}(x|\theta, \mathcal{D})$, on $\Theta \times \mathcal{X}$, where $\tilde{f}(x|\theta, \mathcal{D})$ is a tractable distribution (up to a normalizing constant that does not depend on θ). The AVM of Møller et al. (2006) generates a Markov chain $\{(\theta_n, X_n),\ n \geq 0\}$ on the joint space $\Theta \times \mathcal{X}$ that can be described as follows. Given $(\theta_n, X_n) = (\theta, x)$, one proposes a new point (θ', Y) from the proposal $Q(\theta, d\theta')p_{\theta'}(dy)$, where Q is a transition kernel on Θ. This proposed value is accepted with the M-H ratio

$$\min\left[1, \frac{\tilde{\pi}(\theta', Y)}{\tilde{\pi}(\theta, x)} \frac{Q(\theta', \theta)}{Q(\theta, \theta')} \frac{f_\theta(x)}{f_{\theta'}(Y)} \frac{Z(\theta')}{Z(\theta)}\right] = \min\left[1, \frac{f_{\theta'}(\mathcal{D})}{f_\theta(\mathcal{D})} \frac{\tilde{f}(Y|\theta', \mathcal{D})}{\tilde{f}(x|\theta, \mathcal{D})} \frac{\mu(\theta')}{\mu(\theta)} \frac{Q(\theta', \theta)}{Q(\theta, \theta')} \frac{f_\theta(x)}{f_{\theta'}(Y)}\right].$$

We see that $Z(\theta)/Z(\theta')$ cancels out. Clearly, the invariant distribution of this M-H algorithm is $\tilde{\pi}(\theta, x|\mathcal{D})$, whose θ-marginal is $\int_x \tilde{\pi}(\theta, x|\mathcal{D})dx = \pi(\theta|\mathcal{D})$.

There is a nice simplification on the AVM introduced by Murray et al. (2006), which obtained a valid Θ-valued MCMC sampler with target distribution $\pi(\cdot|\mathcal{D})$. Given θ_n, one generates θ' from $Q(\theta_n, d\theta')$ and Y from $p_{\theta'}(dy)$. Then θ' is accepted with propability

$$\alpha(\theta, \theta', Y) = \min\left[1, \frac{f_{\theta'}(D)\mu(\theta')Q(\theta', \theta)}{f_\theta(D)\mu(\theta)Q(\theta, \theta')} \frac{f_\theta(Y)}{f_{\theta'}(Y)}\right]. \tag{4}$$

The auxiliary variable Y is used to estimate the ratio $Z(\theta)/Z(\theta')$ by a one-sample importance sampling estimator,

$$\frac{Z(\theta)}{Z(\theta')} = \mathbb{E}_{\theta'}\left[\frac{f_\theta(Y)}{f_{\theta'}(Y)}\right] \approx \frac{f_\theta(Y)}{f_{\theta'}(Y)}.$$

Y is discarded after that. Although not a M-H algorithm, Murray et al. (2006) showed that this algorithm (called single variable exchange algorithm or SVEA) generates a Markov chain $\{\theta_n,\ n \geq 0\}$ on Θ with a transition kernel that is reversible (and thus invariant) wrt $\pi(\cdot|\mathcal{D})$. We will denote by T_0 the transition kernel of SVEA:

$$T_0(\theta, A) = \int_{A\times\mathcal{X}} \alpha(\theta, \theta', x)Q(\theta, d\theta')p_{\theta'}(dx) + \mathbf{1}_A(\theta) \int_{\Theta\times\mathcal{X}} [1 - \alpha(\theta, \theta', x)]Q(\theta, d\theta')p_{\theta'}(dx).$$

But as in the case of AVM, this algorithm also requires an exact simulation from $p_{\theta'}(\cdot)$, which in many models is impossible or can be excruciatingly slow to achieve.

We develop an approximate MCMC sampler for $\pi(\cdot|\mathcal{D})$, which does not require exact sampling. The basic idea is to replace the exact sampling from $p_{\theta'}$ by approximately sampling from a distribution close to $p_{\theta'}$.

We phrase this idea in a more general framework. For $\theta, \theta' \in \Theta$ and a positive integer κ, let $P_{\kappa,\theta,\theta'}$ be a transition kernel on $\mathcal{X}$ that depends measurably on θ, θ', s.t. for some measurable function $B_\kappa(x, \theta, \theta')$,

$$\|P_{\kappa,\theta,\theta'}(x, \cdot) - p_{\theta'}(\cdot)\|_{\text{TV}} \leq B_\kappa(x, \theta, \theta').$$

We will naturally impose that $B_\kappa(x,\theta,\theta')$ converges to zero as $\kappa \to \infty$. In the above equation, the total variation distance between two (finite state space) probability mass functions is $\|\mu - \nu\|_{\mathrm{TV}} = \sum_{x\in\mathcal{X}} |\mu(x) - \nu(x)|$. We consider the following algorithm that generates a Markov chain $\{(\theta_n, X_n),\ n \geq 0\}$ on $\Theta \times \mathcal{X}$.

Algorithm 2.1. *At time* 0*, choose* $\theta_0 \in \Theta$ *and* $X_0 \in \mathcal{X}$.
At time n*, given* $(\theta_n, X_n) = (\theta, x)$*:*

1. *Generate* $\theta' \sim Q(\theta_n, \cdot)$.
2. *Set* $Y_0 = x$*. Generate* $Y_\kappa \sim P_{\kappa,\theta,\theta'}(Y_0, \cdot)$*. Compute*

$$\alpha(\theta,\theta',Y_\kappa) = \min\left[1, \frac{f_{\theta'}(\mathcal{D})\mu(\theta')Q(\theta',\theta_n)}{f_{\theta_n}(\mathcal{D})\mu(\theta_n)Q(\theta_n,\theta')}\frac{f_\theta(Y_\kappa)}{f_{\theta'}(Y_\kappa)}\right].$$

3. *With probability* $\alpha(\theta,\theta',Y_\kappa)$*, set* $(\theta_{n+1}, X_{n+1}) = (\theta', Y_\kappa)$*; with probability* $1-\alpha(\theta,\theta',Y_\kappa)$*, set* $(\theta_{n+1}, X_{n+1}) = (\theta, x)$.

Although this algorithm generates a Markov chain on $\mathcal{X} \times \Theta$, we are only interested in the marginal chain $\{\theta_n, n \geq 1\}$. Typically, at the end of the simulation, $\{X_n, n \geq 1\}$ is discarded.

The transition kernel of our approximate MCMC algorithm is $\bar{T}_\kappa$ given by

$$\begin{aligned}\bar{T}_\kappa\Big((\theta,x), A\times B\Big) &= \int_{A\times B} \alpha(\theta,\theta',y_\kappa)Q(\theta,d\theta')P_{\kappa,\theta,\theta'}(x,dy_\kappa)\\ &\quad + \mathbf{1}_{A\times B}(\theta,x)\int_{\Theta\times\mathcal{X}}[1-\alpha(\theta,\theta',y_\kappa)]Q(\theta,d\theta')P_{\kappa,\theta,\theta'}(x,dy_\kappa),\end{aligned}$$

where $\mathbf{1}_A$ is an indicator function of the set A. The limiting kernel of the Markov kernel implemented by Algorithm 2.1 is $\bar{T}_\star$ defined on $\Theta\times\mathcal{X}$ as

$$\begin{aligned}\bar{T}_\star\Big((\theta,x), A\times B\Big) &= \int_{A\times B} \alpha(\theta,\theta',y)Q(\theta,d\theta')p_{\theta'}(dy)\\ &\quad + \mathbf{1}_{A\times B}(\theta,x)\int_{\Theta\times\mathcal{X}}[1-\alpha(\theta,\theta',y)]Q(\theta,d\theta')p_{\theta'}(dy). \qquad (5)\end{aligned}$$

Notice the similarity between $\bar{T}_\star$ and T_0 (the transition kernel of SVEA). $\bar{T}_\star$ is also very closely related to the AVM when $\tilde{f}(x|\theta,\mathcal{D}) = Z^{-1}(\theta)f_\theta(x)$, although this choice does not lead to a tractable algorithm. We have the following result which shows that the limiting kernel $\bar{T}_\star$ will sample correctly from the posterior distribution. Since $\bar{T}_\kappa$ is close to $\bar{T}_\star$ for large κ, this implies that Algorithm 2.1 approximately sample from the posterior distribution. We make this intuition rigorous below.

Proposition 2.1. *Suppose that the posterior distribution* $\pi(\cdot|\mathcal{D})$ *is the unique invariant distribution of* T_0*, and that* $\bar{T}_\star$ *admits an invariant distribution with density* $\bar{\pi}_\star$ *on* $\Theta\times\mathcal{X}$*. Then* $\int \bar{\pi}_\star(x,\theta)dx = \pi(\theta|\mathcal{D})$*. That is, the* θ*-marginal of* $\bar{\pi}_\star$ *is* $\pi(\cdot|\mathcal{D})$*.*

Proof. Let $\pi_\star$ be the θ-marginal of $\bar{\pi}_\star$. That is, $\pi_\star(\theta) = \int \bar{\pi}_\star(\theta, x)dx$. Notice that from the definitions, $\bar{T}_\star((\theta, x), A \times \mathcal{X}) = T_0(\theta, A)$. Now let $\{(\theta_n, X_n),\ n \geq 0\}$ be a Markov chain with transition kernel $\bar{T}_\star$ in stationarity. Therefore $(\theta_{n-1}, X_{n-1}) \sim \bar{\pi}_\star$ and for any measurable subset A of Θ,

$$\begin{aligned}
\pi_\star(A) = \mathbb{P}\left(\theta_n \in A, X_n \in \mathcal{X}\right) &= \int \bar{\pi}_\star(d\theta, dx)\bar{T}_\star\left((\theta, x), A \times \mathcal{X}\right) = \int \bar{\pi}_\star(d\theta, dx)T_0(\theta, A) \\
&= \int \mathbb{P}\left(\theta_{n-1} \in d\theta\right) T_0(\theta, A) = \int \pi_\star(d\theta)T_0(\theta, A).
\end{aligned}$$

In other words, $\pi_\star$ is also an invariant distribution for T_0. The result then follows easily from the assumptions. □

It is known (Murray et al., 2006) that the kernel T_0 has invariant distribution $\pi(\cdot|\mathcal{D})$, and in practice it is usually easy to construct T_0 such that it has a unique invariant distribution. We will see below that under some regularity conditions, $\bar{T}_\star$ admits an invariant distribution $\bar{\pi}_\star$. Thus in many applications the conclusion of Proposition 2.1 holds.

2.0.1. Building the kernel $P_{\kappa,\theta,\theta'}$

It remains to describe how we build the approximating kernel $P_{\kappa,\theta,\theta'}$. A natural candidate is

$$P_{\kappa,\theta,\theta'}(x, \cdot) \stackrel{\text{def}}{=} P_{\theta'}^{\kappa}(x, \cdot), \tag{6}$$

for some integer κ, where $P_{\theta'}$ is a Markov kernel with invariant distribution $p_{\theta'}$. We refer to this choice of $P_{\kappa,\theta,\theta'}$ as the homogeneous instrument kernel.

If we assume that $X_n \sim p_\theta$ and p_θ and $p_{\theta'}$ are far apart, instead of κ iterations from $P_{\theta'}$ starting from X_n, a possibly useful alternative is to build a bridge of distributions between p_θ and $p_{\theta'}$. Let $\{f_t,\ 0 \leq t \leq 1\}$ be a smooth path of un-normalized densities between f_θ and $f_{\theta'}$ such that $f_0 = f_\theta$ and $f_1 = f_{\theta'}$. Define $Z(t) = \int_{\mathcal{X}} f_t(x)dx$ to be the normalizing constant for f_t, $p_t(\cdot) = f_t(\cdot)/Z(t)$, and P_t a transition kernel (e.g., M-H kernel) with invariant distribution p_t. We then pick κ intermediate distributions between p_θ and $p_{\theta'}$ by choosing $0 \leq t_1 \ldots \leq t_\kappa \leq 1$. Step 2 of Algorithm 2.1 is then implemented as follows. Given $Y_\ell, \ell = 0, \ldots, \kappa - 1$, we generate $Y_{\ell+1}$ from $P_{t_\ell+1}(Y_\ell, \cdot)$. So the transition kernel from x to y_κ is

$$P_{\kappa,\theta,\theta'}(x, \cdot) \stackrel{\text{def}}{=} \int_{\mathcal{X}} P_{t_1}(x, dy_1) \ldots \int_{\mathcal{X}} P_{t_{\kappa-1}}\left(y_{t_{\kappa-2}}, dy_{t_{\kappa-1}}\right) P_{t_\kappa}\left(y_{t_{\kappa-1}}, \cdot\right). \tag{7}$$

For large κ, the intermediate distributions will be very close to each other, making the transition from one distribution to another easier. With properly chosen $t_1, \ldots, t_\kappa$, as κ increases, we expect

Y_κ to converge to $p_{\theta'}$. We refer to this choice of $P_{\kappa,\theta,\theta'}$ as the nonhomogeneous instrument kernel, as each P_{t_ℓ} $(\ell = 1, \ldots, \kappa)$ is different.

One generic choice of path is the geometric path defined as $f_t(x) = f_\theta^{1-t}(x) f_{\theta'}^t(x)$. Throughout the paper, we will use this path for the nonhomogeneous instrument kernel. For an ERG model in its canonical form, path defined through $\theta(t) = \theta + t(\theta' - \theta)$ with $f_t(x) = f_{\theta(t)}(x)$ is equivalent to the geometric path.

In the simulations that we have performed, the nonhomogeneous instrument kernel does not show any particular advantage over the homogeneous one. This might be due to the fact that the conditional distribution of X_n given (X_{n-1}, θ_{n-1}) in Algorithm 2.1 is not $p_{\theta_{n-1}}(\cdot)$. Nevertheless, we found the bridging idea interesting and it might be useful for some other models.

2.0.2. Choosing the parameter κ

It is possible to run Algorithm 2.1 with κ held fixed. In this case, the invariant distribution of the algorithm will not be precisely the posterior distribution $\pi(\cdot|\mathcal{D})$. The larger κ, the longer the computing time, and the closer to $\pi(\cdot|\mathcal{D})$ the corresponding invariant distribution gets. The total variation norm between the two distributions is bounded in Theorem 2.1. From experience, we found that κ in the range $100 - 200$ yields a reasonable approximating distribution for problems of the size of those considered in this paper.

On the other hand, if more computing effort is allowed, we can make κ increase with n, using $\kappa = \kappa_n$ at the n-th iteration of Algorithm 2.1. For instance, $\kappa_n = \lceil d \log(n+1) \rceil$ or $\kappa_n = \lceil d\sqrt{n} \rceil$, where $\lceil a \rceil$ is the smallest integer that is no less than a. The advantages is intuitively clear but we also show in Theorem 2.2 that with increasing κ, the limiting distribution of θ_n is precisely $\pi(\cdot|\mathcal{D})$. Furthermore, a strong law of large numbers and central limit theorem also hold. Of course, these limiting arguments with $\kappa_n \to \infty$ come at the expense of greater computing effort. This is nevertheless the approach taken below in our numeric examples, where we use $\kappa_n = \lceil d \log(1+n) \rceil$, with d mostly in the range of $10 - 20$.

2.1. Theory

We now give some theoretical justification of the method. First, we define some terminology and notations to be used in the entire section. For a transition kernel P, we denote by P^n, $n \geq 0$, its n-th iterate, with $P^0(x, A) = \mathbf{1}_A(x)$. The total variation normal between two measures λ and μ is defined as $\|\lambda - \mu\|_{\mathrm{TV}} \stackrel{\text{def}}{=} \frac{1}{2} \sup_{\{|h| \leq 1\}} |\lambda(h) - \mu(h)|$, where h is any measurable function. $Ph(x) = \int P(x, y) h(y) dy$, and $\mu h = \int \mu(x) h(x) dx$. Let $\lfloor a \rfloor$ be the greatest integer not exceeding a.

For simplicity, we assume that Θ is a compact subset of a p-dimensional Euclidean space $\mathbb{R}^p$ equipped with its Borel σ-algebra. We recall also that the sample space $\mathcal{X}$ is finite. Furthermore, we introduce the following assumptions.

A1 There exist $\epsilon_\kappa > 0$ and a positive integer n_κ, s.t. for all $(\theta, x) \in \Theta \times \mathcal{X}$, and for all $(\theta_1, x_1), \ldots, (\theta_{n_\kappa}, x_{n_\kappa}) \in \Theta \times \mathcal{X}$,

$$P_{\kappa,\theta,\theta_1}(x, x_1) \cdots P_{\kappa,\theta_{n_\kappa - 1},\theta_{n_\kappa}}(x_{n_\kappa - 1}, x_{n_\kappa}) \geq \epsilon_\kappa. \tag{8}$$

The compactness of Θ makes it possible to check A1 for many examples. See Proposition 2.2-2.3 below.

A2 $B_\kappa(x, \theta, \theta')$ does not depend on x, and denote it by $B_\kappa(\theta, \theta')$. Moreover,

$$\sup_{\theta \in \Theta} \int_\Theta Q(\theta, \theta') B_\kappa(\theta, \theta') \to 0, \quad \text{as } \kappa \to \infty.$$

Theorem 2.1. *Assume A1-A2 and suppose also that Q, μ and f_θ are positive and continuous functions of θ (and θ' in Q). Then $\bar{T}_\kappa$ and $\bar{T}_\star$ have unique invariant distributions denoted by $\bar{\pi}_\kappa$ and $\bar{\pi}_\star$ respectively. Furthermore,*

$$\|\bar{\pi}_\kappa - \bar{\pi}_\star\|_{\mathrm{TV}} \leq C \sup_{\theta \in \Theta} \int_\Theta Q(\theta, \theta') B_\kappa(\theta, \theta') d\theta', \tag{9}$$

where C is some constant that does not depend on θ or κ.

Proof. The basic idea is to bound the distance between the $\bar{T}_\kappa$ and $\bar{T}_\star$. This distance depends on the total variation norm between $P_{\kappa,\theta,\theta'}$ and $p_{\theta'}$, which is bounded by $B_\kappa(\theta, \theta')$. See section Appendix A for a detailed proof. □

Let us denote $\pi_\kappa(\cdot) \stackrel{\text{def}}{=} \bar{\pi}_\kappa(\cdot \times \mathcal{X})$ the θ-marginal of $\bar{\pi}_\kappa$. Under the assumptions of the Theorem, it is straightforward to check that the transition kernel T_0 of the SVEA algorithm of (Murray et al., 2006) has $\pi(\cdot|\mathcal{D})$ as unique invariant distribution. Therefore by Proposition 2.1 and the bound (9), we conclude that

$$\|\pi_\kappa(\cdot) - \pi(\cdot|\mathcal{D})\|_{\mathrm{TV}} \leq C \sup_{\theta \in \Theta} \int_\Theta Q(\theta, \theta') B_\kappa(\theta, \theta') d\theta'. \tag{10}$$

Remark 2.1. *The bound (10) tells us that if we run Algorithm 2.1 under the conditions of the theorem with κ held fix, then the marginal distribution of θ_n will converge to the distribution π_κ, which is within $C\sup_{\theta\in\Theta}\int_\Theta Q(\theta,\theta')B_\kappa(\theta,\theta')d\theta'$ of the posterior distribution.*

Suppose now that we run Algorithm 2.1 using $\kappa=\kappa_n$ at the n-th iteration for some nondecreasing sequence $\{\kappa_n\}$, s.t. $\kappa_n\to\infty$. Since the transition kernel now is $\bar{T}_{\kappa_n}$, different for each n, then Algorithm 2.1 generates a nonhomogeneous Markov chain $\{(\theta_n,X_n),\ n\geq 0\}$. Given $\mathcal{F}_n=\sigma\{(\theta_0,X_0),\ldots,(\theta_n,X_n)\}$, the conditional distribution of (θ_{n+1},X_{n+1}) is $\bar{T}_{\kappa_n}\Big((\theta_n,X_n),A\times B\Big)$. We can say the following about this process.

Theorem 2.2. *Suppose that in Algorithm 2.1, we set $\kappa=\kappa_n$ and let $\{(\theta_n,X_n),\ n\geq 0\}$ be the resulting nonhomogeneous Markov chain. Assume A1-A2. Suppose also that $\inf_{\kappa\geq 1}\epsilon_\kappa>0$, and $\sup_{\kappa\geq 1}n_\kappa<\infty$. Denote the distribution of (θ_n,X_n) by $\mathscr{L}_{(\theta_n,X_n)}$. We have:*

(1)

$$\lim_{n\to\infty}\left\|\mathscr{L}_{(\theta_n,X_n)}-\bar{\pi}_\star\right\|_{\mathrm{TV}}=0. \tag{11}$$

(2) For any measurable function $h:\Theta\times\mathcal{X}\to\mathbb{R}$,

$$\lim_{n\to\infty}n^{-1}\sum_{i=1}^{n}h(\theta_i,X_i)=\bar{\pi}_\star h,\quad a.s., \tag{12}$$

where $\bar{\pi}_\star h=\int_\Theta\int_{\mathcal{X}}\bar{\pi}_\star(\theta,x)h(\theta,x)d\theta dx$. Furthermore, if

$$\sum_{i=1}^{\infty}i^{-1/2}\int_\Theta Q(\theta,\theta')B_{\kappa_i}(\theta,\theta')d\theta'<\infty,\ \textit{then} \tag{13}$$

$$n^{-1/2}\sum_{i=1}^{n}\Big(h(\theta_i,X_i)-\bar{\pi}_\star h\Big)\xrightarrow{\mathcal{D}}N\Big(0,\sigma^2(h)\Big),\quad \textit{as } n\to\infty, \tag{14}$$

where

$$\sigma^2(h)=Var\Big(h(\theta_0,X_0)\Big)+2\sum_{i\geq 1}Cov\Big(h(\theta_0,X_0),h(\theta_i,X_i)\Big),$$

where the Var and Cov are computed under the assumption that $\{(\theta_n,X_n),n\geq 0\}$ is a stationary Markov chain with kernel $\bar{T}_\star$ and invariant distribution $\bar{\pi}_\star$.

Proof. See section Appendix B for a detailed proof. □

By marginalization, the above theorem implies the following about the marginal process $\{\theta_n,\ n\geq 1\}$ generated from Algorithm 2.1 with $\kappa=\kappa_n$: under the above assumptions, $\lim_{n\to\infty}\|\mathscr{L}_{\theta_n}-\pi_\star\|_{\mathrm{TV}}=0$; $\lim_{n\to\infty}n^{-1}\sum_{i=1}^{n}h(\theta_i)=\pi_\star h$, *a.s.*; and $n^{-1/2}\sum_{i=1}^{n}\left(h(\theta_i)-\pi_\star h\right)\xrightarrow{\mathcal{D}}N\left(0,\sigma^2(h)\right)$, *as* $n\to\infty$, where $\sigma^2(h)$ is as in Theorem 2.2 with $h(\theta,x)=h(\theta)$.

Let us now show that the assumptions of the above theorems hold for the ERG model.

Proposition 2.2. *For an ERG model in the form of (1) for a binary symmetric network, if the parameter space is compact, $P_{\kappa,\theta,\theta'}$ in Algorithm 2.1 is the homogeneous instrument kernel (6), and $P_{\theta'}$ is the random sampler, then Theorems 2.1 and 2.2 hold.*

Proof. Please refer to section 3 for a detailed description of random sampler. Given x, let $N(x) \stackrel{\text{def}}{=} \{y :$ triangular matrices y and x differ by one and only one entry$\}$. Assume that the network consists of n people, so there are $m = n(n-1)/2$ entries. For any $\theta \in \Theta$, the kernel of the random sampler is $P_\theta(x,y) = \frac{1}{m}\min[1, \frac{f_\theta(y)}{f_\theta(x)}]$ for $y \in N(x)$, and is 0 for all other $y \neq x$; $P_\theta(x,x) = 1 - \sum_{y\in N(x)} \frac{1}{m}\min[1, \frac{f_\theta(y)}{f_\theta(x)}]$. By the compactness of Θ, there exists $\epsilon_0 > 0$ such that for any $x \in \mathcal{X}$, any $y \in N(x)$, and any $\theta \in \Theta$, $P_\theta(x,y) \geq \epsilon_0$. For any $x, y \in \mathcal{X}$, one can change x to y by toggling only one dyad at a time in a maximum of m moves. Thus $P_\theta^m(x,y) \geq \epsilon_0^m$ for all $x, y \in \mathcal{X}$ and all $\theta \in \Theta$.

With the same idea, we can show that $P_{\kappa,\theta,\theta'} = P_{\theta'}^\kappa$ satisfies A1 with $n_\kappa = \lceil m/\kappa \rceil$, $\epsilon_\kappa = \epsilon_0^{n_\kappa \kappa}$. It follows directly that $\sup_\kappa n_\kappa \leq m < \infty$ and $\inf_\kappa \epsilon_\kappa = \epsilon_0^{2m-2} > 0$. Recall $B_\kappa(x,\theta,\theta') = \|P_{\theta'}^\kappa - p_{\theta'}\|_{\text{TV}}$. By Meyn and Tweedie (2003) Theorem 16.02, $B_\kappa(x,\theta,\theta') \leq (1-\epsilon_0)^{\lfloor \kappa/m \rfloor}$, which does not depend on x. A2 and (13) can be then verified. All other conditions are trivial to check. □

Proposition 2.3. *For an ERG model in the form of (1) for a binary symmetric network, if the parameter space is compact, $P_{\kappa,\theta,\theta'}$ in Algorithm 2.1 is the nonhomogeneous instrument kernel (7), and both P_θ and $P_{\theta'}$ are the random sampler, then Theorems 2.1 and 2.2 hold, except for the central limit theorem result (14). If $\sum_{n\geq 1} n^{-1/2}\kappa_n^{-1} < \infty$ then (14) holds as well.*

Proof. The same arguments as above show that A1 holds; in fact with the same constants ϵ_κ and n_κ. Deriving the bound $B_\kappa(x,\theta,\theta')$ requires some work. We have

$$\begin{aligned}
B_\kappa(x,\theta,\theta') &= \frac{1}{2}\sup_{|h(x)|\leq 1}\left|P_{t_1}\ldots P_{t_{\kappa-1}}P_{\theta'}h - p_{\theta'}h\right| \\
&\leq \frac{1}{2}\sup_{|h|\leq 1}\left|\sum_{j=0}^{\kappa-1} P_{t_1}\ldots P_{t_j}(P_{t_{j+1}} - P_{\theta'})(P_{\theta'}^{\kappa-j-1} - p_{\theta'})h\right| + \|P_{\theta'}^\kappa - p_{\theta'}\|_{\text{TV}} \\
&\leq \frac{1}{\kappa}\sum_{i=0}^{\kappa-1} i(1-\epsilon_0 m)^{\lfloor i/m \rfloor} + (1-\epsilon_0 m)^{\lfloor \kappa/m \rfloor}.
\end{aligned}$$

Thus for large κ, $B_\kappa(x,\theta,\theta') = O(1/\kappa)$ and A2 holds. And (13) is valid if $\sum_{n\geq 1} n^{-1/2}\kappa_n^{-1} < \infty$. □

Remark 2.2. *While the idea of bridging between p_θ and $p_{\theta'}$ is appealing, it is theoretically challenging to obtain a good idea of $B_\kappa(x, \theta, \theta')$. As shown above, $B_\kappa(x, \theta, \theta') = O(1/\kappa)$. However, the bound is quite loose, and we believe that the chain converges at a much faster rate.*

3. Metropolis-Hastings moves on spaces of large networks

In this section, we restrict attention to the ERG model (1). Efficient sampling from the ERG distribution p_θ is important for successfully fitting the ERG model to data. For example, our approximate MCMC algorithm as applied to the ERG model is sensitive to the mixing of the $\mathcal{X}$-move instrument kernel $P_{\kappa,\theta,\theta'}$. For the two instrument kernels specified in section 2.0.1, their mixing depends further on that of $P_{\theta'}$ and P_{t_ℓ}, respectively. When the random sampler is used to build these kernels, the resulting algorithm does not perform well on large sparse networks. As mentioned in the introduction, the same issue arises in computing the MLE for the model.

Random sampler works as follows. Given the current network y, we choose two nodes uniformly at random, and toggle their connection status (i.e. break the edge if two people are connected, otherwise build an edge). All other edges remain the same. This new network y' which differs only by one edge from y, is accepted with probability $\min[1, \frac{f(y')}{f(y)}]$, where $f(y)$ is the un-normalized density of an ERG model, and we ignore parameters here for notation lucidity.

As discussed in the introduction, its mixing can be slow due to the sparsity of the real networks in practice. Most dyads in sparse networks are disconnected. So for random sampler, there is a high probability that an empty dyad will be chosen to toggle. But the parameter controlling network density is usually negative for sparse networks, which results in a high chance of rejecting the proposal. A refined kernel is TNT (Morris et al., 2008), where with probability .5 an existing edge is selected to toggle, and with probability .5 a disconnected dyad is selected to toggle. Then we accept it with probability $\min[1, \frac{f(y')}{f(y)}]$. This way we are able to explore the network space more efficiently.

When the model involves statistics that depict higher order structures (as opposed to nodal or dyadic structure) such as triangles, simple proposals may be inefficient for large networks, in the sense that the statistic value will not vary easily. For instance, in a sparse large network, it will take many steps of toggling dyads before a triangle is formed.

Transitivity is an important characteristic to study in social network modeling, as it captures the idea that people having common friends tent to become friends. It is often quantified by statistics involving triangles, e.g., the number of triangles and alternating k-triangle. Recent studies (Snijders, 2002; Handcock, 2003) found that some specifications of the ERG model will lead to degenerate distributions, meaning that the distributions put most of the probability mass on a small subset of the sample space. The number of triangles could contribute to model degeneracy in some cases. Alternating k-triangle statistic (Snijders et al., 2006) was introduced to prevent model degeneracy. For a network of n people, we introduce its analytical expression given in Hunter and Handcock (2006),

$$v(x;\theta) = e^{\theta} \sum_{k=1}^{n-2} \{1 - (1 - e^{-\theta})^k\} D_k(x), \tag{15}$$

where $D_k(x)$ is the number of connected dyads that share exactly k neighbors in common. The intuition is that as k increases, we less favor the formation of k-triangles (a k-triangle consists of k triangles that share one common edge) to prevent the edge explosion.

Neither random sampler nor TNT can help form triangles efficiently. In both proposals, a dyad is selected uniformly at random. For sparse networks, most of the time, a triangle is built from three disconnected nodes. Then all three pairwise dyads need to be chosen, which will take a long time. An easier way is to choose an open triangle (or two-star) and complete the missing edge. Here we propose a new M-H kernel based on this idea. Notice that slow mixing does not only exist for models with triangles, but also with many higher order structures, which are hard to form or/and destroy in sparse networks. We take triangles as an example for its popularity in social network modeling. As long as readers are aware of the problem, it is not a difficult task to modify existing proposals accordingly.

3.1. OTNT proposal

The kernel is built on TNT, and we take one step further to complete triangles with the help of k-stars ($k \geq 2$). We assume the network is an n by n binary symmetric network, but the same idea can be carried over to more complicated networks. Given the current network y, we propose a new y' by adding or deleting an edge as follows. With probability w_1 we pick a disconnected dyad uniformly at random and propose adding the edge between them. With probability w_2, we randomly pick a connected dyad and propose breaking the connection. The rest of the time, we randomly choose a node of degree at least two, i.e., k-star ($k \geq 2$), and then randomly select two of its neighbors. If the three nodes form an open triangle in y, i.e. the neighbors are disconnected, then we propose connecting them to form the triangle. Otherwise we do not change the network. The basic idea is to complete open triangles, and hence we name this kernel OTNT (open triangle-tie-no tie). When $w_1 + w_2 = 1$, it turns back to TNT.

We accept y' with probability $\min[1, \frac{f(y')q(y',y)}{f(y)q(y,y')}]$, where q is the transition kernel of OTNT. For the two instrument kernels in section 2.0.1, $f(y)$ refers to $f_{\theta'}$ and f_{t_ℓ}, respectively. $q(y,y') \stackrel{\text{def}}{=} w_1 q_1(y,y') + w_2 q_2(y,y') + (1-w_1-w_2)q_3(y,y')$. Define $y'-y=1$, if y' has one more edge than y and they only differ by this edge. We have

$$q_1(y,y') = \frac{1}{\frac{n(n-1)}{2} - \mathrm{E}(y)}, \quad \text{if } y'-y=1, \text{ and 0 otherwise.}$$

$$q_2(y,y') = \frac{1}{\mathrm{E}(y)}, \quad \text{if } y-y'=1, \text{ and 0 otherwise.}$$

$$q_3(y,y') = \frac{1}{m}\sum_{k=1}^{l}\frac{1}{\binom{n_k}{2}}, \quad \text{if } y'-y=1,$$

where n is the number of nodes, $E(y)$ is the number of edges in the network, and m is the number of k-stars ($k \geq 2$) in y. For q_3, assuming y and y' differ by edge (i,j), then l is the number of common neighbors i and j have, and n_k is the degree of their k-th common neighbor. For q_3, if $y'=y$, we always accept y' and do not need to calculate $q_3(y,y')$ nor $q(y,y')$. For all other y', $q_3(y,y')=0$.

In section 4, we will compare the performance of random sampler, TNT and OTNT on a large social network example, and the result is in favor of OTNT. The key point is to form triangles more often with structures that can be tracked easily, such as k-stars in OTNT.

4. Numerical Examples

In this section, we showcase the efficiency and accuracy of our approximate MCMC algorithms in Ising model and conditional random field. Although these models are not ERG models, they both suffer from the intractable normalizing constant problem. We also apply the algorithm to a large social network and compare the performance of random sampler, TNT and OTNT. In all the examples, both homogeneous and nonhomogeneous instrument kernels introduced in section 2.0.1 perform well. There is negligible difference in computing and programming effort between them, so we just report the computing time for the first one. Unless otherwise noted, we choose $\kappa_n = \lceil d\log(n+1)\rceil$ at time n of Algorithm 2.1 with d a fixed value.

4.1. Ising model

We have a lattice with N^2 nodes. Each node $x_{ij}, 1 \leq i,j \leq N$ is connected with its horizontal and vertical neighbors, and is a $\{-1,1\}$-valued random variable. Let $x = (X_{ij})$ be an $N \times N$

matrix. Assume that these random variables jointly have an Ising distribution with probability mass function

$$p_\theta(x) = \exp\left\{\theta\left(\sum_{i=1}^{N}\sum_{j=1}^{N-1} x_{ij}x_{i,j+1} + \sum_{i=1}^{N-1}\sum_{j=1}^{N} x_{ij}x_{i+1,j}\right)\right\}/Z(\theta). \tag{16}$$

The normalizing constant cannot be computed unless the size of N^2 is small. This type of model is fairly common in image analysis.

We simulate the data by perfect sampling using Propp-Wilson algorithm (Propp and Wilson, 1996) with $\theta = 0.25$ for $N = 20, 50, 100$. The prior is $\mu(\theta) \sim \text{Uniform}(0,1)$, and the posterior is

$$\pi(\theta|x) \propto \exp\left\{\theta\left(\sum_{i=1}^{N}\sum_{j=1}^{N-1} x_{ij}x_{i,j+1} + \sum_{i=1}^{N-1}\sum_{j=1}^{N} x_{ij}x_{i+1,j}\right)\right\}\mathbf{1}_{(0,1)}(\theta).$$

At time n of Algorithm 2.1, we propose θ' using a Random Walk Metropolis sampler with proposal distribution $N(\theta_n, \sigma^2)$ ($\sigma = .05$ for $N = 20$, $\sigma = .01$ for $N = 50, 100$). The random sampler is used to sample the networks in Step 2 of Algorithm 2.1. For the nonhomogeneous instrument kernel, $f_\ell = f_\theta^{1-\ell/\kappa} f_{\theta'}^{\ell/\kappa}$, for $\ell = 1, \ldots, \kappa$.

The goal of this simulation is twofold. Firstly, we implement Algorithm 2.1 with different fixed values of κ, and examine how far away π_κ is from $\pi(\cdot|x)$. Secondly, we take $\kappa = \kappa_n = \lceil d\log(1+n)\rceil$ and see how well the algorithm performs compared to the AVM. For the comparisons, we need to obtain samples from the true posterior distribution $\pi(\cdot|x)$. We do this by running Algorithm 2.1 for a long time with increasing κ_n (at a fast rate). We use the homogeneous instrument kernel with $\kappa_n = \lceil 100\sqrt{n}\rceil$, and run the algorithm for 100,000 iterations.

We run Algorithm 2.1 for $100,000$ iterations for different values of κ. We use the output to estimate the Kolmogorov-Smirnov distance between π_κ and $\pi(\cdot|x)$ defined as

$$\mathsf{KS}\left(\pi_\kappa, \pi(\cdot|x)\right) = \sup_{\theta\in[0,1]}\left|\int_0^\theta \pi_\kappa(u)du - \int_0^\theta \pi(u|x)du\right|.$$

In calculating the KS distance, we discard the first 20% of the outcomes as burn-in and a certain thinning (100 for $N = 20, 50$ and 200 for $N = 100$) is applied to the remaining data to reduce the auto-correlation. The results are presented in table 1. As expected from Theorem 2.1, the KS distance between π_κ and $\pi(\cdot|x)$ decreases as κ increases. The drop in the KS distance is slower for larger N as expected. Also the homogeneous and nonhomogeneous instrument kernels give similar KS distances.

Table 1: Ising model example: Kolmogorov-Smirnov distance between π_κ and $\pi_\star$ for different size of the lattice, $N^2 = 20^2, 50^2, 100^2$. The homogeneous and nonhomogeneous instrument kernels are differentiated by subscripts 'h' and 'n', respectively.

	20^2		50^2		100^2	
κ	KS_h	KS_n	KS_h	KS_n	KS_h	KS_n
10	.42	.44	.44	.45	.56	.56
30	.31	.32	.34	.36	.47	.49
50	.28	.30	.29	.34	.47	.46
80	.24	.23	.26	.29	.42	.44
100	.20	.19	.27	.26	.44	.45
300	.15	.17	.21	.20	.31	.33
500	.07	.10	.14	.17	.33	.32
800	.07	.07	.13	.15	.26	.28
1000	.05	.07	.11	.14	.28	.28

One thing worth mentioning is that the magnitude of σ^2 in proposing θ' affects the algorithm nonnegligibly when κ is small. The uniform bound of $B_\kappa(x, \theta, \theta')$ we get in the theory section is loose for small κ. A tighter bound depends on if x is from a distribution close to $p_{\theta'}$, as well as the distance between θ and θ' (for the nonhomogeneous instrument kernel). Large σ^2 thus leads to a large $B_\kappa(x, \theta, \theta')$. We use a smaller σ for $N = 50$ than $N = 20$. It is then not surprising to observe almost no difference in KS distance between them for small κ. When κ gets large, π_κ is closer to $\pi_\star$ for $N = 20$ than $N = 50$.

Now we allow κ to increase with n ($\kappa_n = \lceil d \log(n + 1) \rceil$) so that the limiting distribution of θ_n is $\pi_\star$. We carry out 10,000 iterations. The first 2000 are discarded, and every 10-th of the remaining points are used to do posterior inference (same in all examples below). For comparison, we also run the AVM of (Møller et al., 2006) in the setting $N = 20$, where θ' is also proposed from $N(\theta_n, .05^2)$. To reduce the computational load of perfect sampling, the prior of the θ is set to be Uniform$(0, .4)$, and the initial value is $\theta = .2$. The 'true' posterior mean of $\pi_\star$ is obtained using the above sample from $\pi_\star$ with burn-in of 20,000 and thinning of 10. The results are listed in table 2.

Again, the performance of both kernels is similar. It is clear from the table that our algorithm recovers the true posterior mean well. As d increases and hence κ_n, our posterior mean will be

Table 2: Ising model example: the posterior mean of the parameter $\bar{\theta}$, its asymptotic standard deviation $\sigma(\theta)$ (taking the auto-correlation into account), the values of d, the proposal standard deviation σ and the computing time for different N^2. The homogeneous and nonhomogeneous instrument kernels are distinguished by subscripts 'h' and 'n', respectively. The computing time is for homogeneous instrument kernel.

N^2	True $\bar{\theta}$	$\bar{\theta}_h$	$\bar{\theta}_n$	$\sigma(\theta)_h$	$\sigma(\theta)_n$	d	σ	Time (mins)
20^2	.252	.256	.253	.037	.013	100	.05	5.03
20^2 (AVM)	.252	.247		.027		NA	.05	577
50^2	.249	.255	.254	.044	.038	5	.01	1.02
50^2	.249	.250	.250	.021	.015	15	.01	1.25
100^2	.252	.254	.255	.031	.026	20	.01	3.26
100^2	.252	.252	.253	.012	.016	100	.01	7.04

closer to the true value, and the posterior standard deviation is smaller. For the same reason as above, larger d or smaller σ is needed as N increases. The computing speed of our algorithm is remarkably faster than AVM.

The trace plots, histograms and lag-50 auto-correlation plots for our algorithm with both instrument kernels and AVM in the case of $N = 20, d = 100, \sigma = .05$ are given in figure 1. Our algorithm converges to the posterior distribution with good mixing. AVM sampler mixes slowly and often gets stuck at large θ, which gives the false peak in the posterior distribution.

4.2. Conditional random field

We have N^2 individuals on the square lattice $\{1, \cdots, N\} \times \{1, \cdots, N\}$. Assume each individual has a vector of p covariates $A_{ij}^T = (A_{ij,1}, \cdots, A_{ij,p})$ and a dependent variable $X_{ij} \in \{-1, 1\}$, $i, j \in \{1, \cdots, N\}$. Further assume that the dependent variables $(X_{11}, X_{12}, \cdots, X_{NN})$ form a conditional random field given $A = (A_{11}, A_{12} \cdots, A_{NN})$ with distribution

$$p_{\beta,\gamma}(x_{11}, x_{12}, \cdots, x_{NN}|A) \propto \exp\left\{\sum_{i=1}^{N}\sum_{j=1}^{N} x_{ij}A_{ij}^T\beta + \gamma_1 \sum_{i=1}^{N}\sum_{j=1}^{N-1} x_{ij}x_{i,j+1} + \gamma_2 \sum_{i=1}^{N-1}\sum_{j=1}^{N} x_{ij}x_{i+1,j}\right\},$$

for parameters $\beta \in \mathbb{R}^p$ and $\gamma_1, \gamma_2 > 0$. Conditional random field models have a wide range of applications, such as speech processing and computational biology.

We simulate the data through perfect sampling, with $\beta = (.1, .3, .5, .7)$, $\gamma_1 = .2$, $\gamma_2 = .5$, and two different $N = 20, 50$. Each A_{ij} is generated from $N(0, I_4)$. We use prior $\beta \sim N(0, I_4)$ and

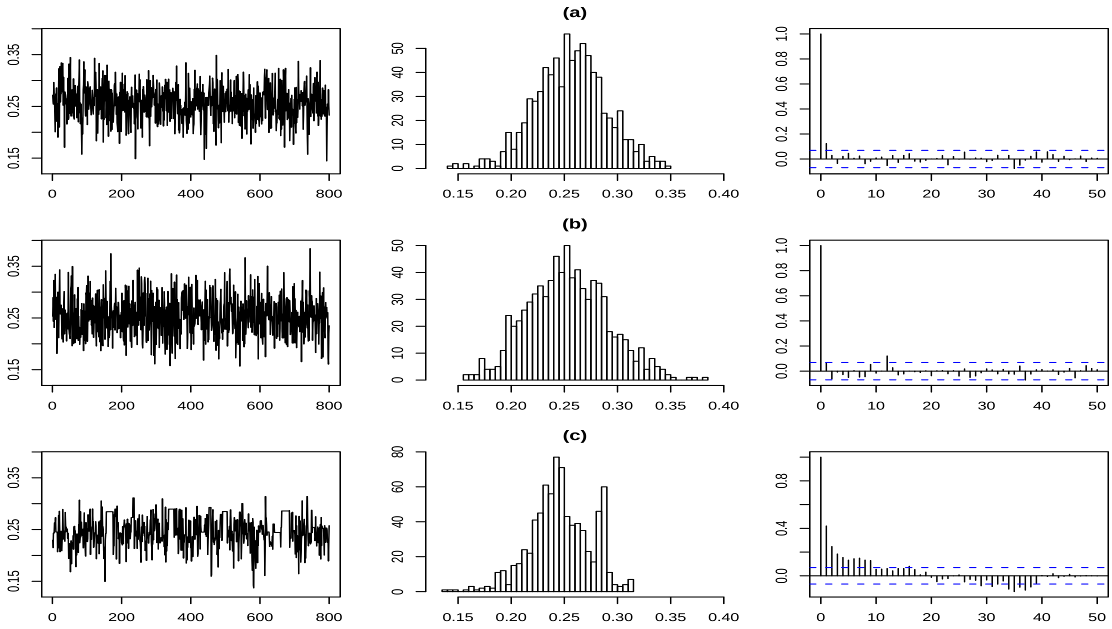

Figure 1: Ising model example: (a) Trace plot, histogram, and auto-correlation plot for Algorithm 2.1 with homogeneous instrument kernel, (b) with nonhomogeneous instrument kernel, (c) for AVM. Data is generated from Ising model with $N^2 = 20^2, \theta = 0.25$.

$\gamma_1, \gamma_2 \sim \mathrm{IG}(1, 1)$. Each parameter is drawn in turn from its full conditional distribution with approximate sampling, and random sampler is used to make a transition from Y_ℓ to $Y_{\ell+1}$. The proposal standard deviations σ for parameters are .05 and .01 for $N = 20, 50$, respectively. In both cases, we use $\kappa_n = \lceil d \log(1+n) \rceil$ where $d = 10$. Results (after burn-in of 2000, and thinning of 10) are shown in table 3. Again, posterior means of our algorithm are very close to the true values obtained in a similar way as in Ising example.

4.3. Real data analysis - large social network modeling

The posterior distributions of some ERG models are doubly intractable (e.g., models with alternating k-triangle statistic), and cannot be handled by exact sampling. The improved efficiency of our algorithm makes it feasible to do inference on large networks in a Bayesian framework.

The Faux Magnolia High school data set (Goodreau et al., 2008) includes a symmetric binary friendship network x of $n = 1461$ students and their demographic information A. We follow the ERG model specification in Goodreau et al. (2008):

$$p_{\theta_1,\theta_2,\beta}(x|A) = \exp\left\{\theta_1 \mathrm{E}(x) + \theta_2 v(x) + \beta^T S(x, A)\right\} / Z(\theta_1, \theta_2, \beta). \tag{17}$$

Table 3: Conditional random field example: the posterior means of the parameters $\bar{\theta}$ and their asymptotic standard deviations $\sigma(\theta)$, (taking the auto-correlation into account) for different N^2. The homogeneous and nonhomogeneous instrument kernels are distinguished by subscripts 'h' and 'n', respectively.

	20^2					50^2				
θ	True $\bar{\theta}$	$\bar{\theta}_h$	$\bar{\theta}_n$	$\sigma(\theta)_h$	$\sigma(\theta)_n$	True $\bar{\theta}$	$\bar{\theta}_h$	$\bar{\theta}_n$	$\sigma(\theta)_h$	$\sigma(\theta)_n$
β_1	.081	.073	.084	.199	.133	.092	.097	.093	.031	.115
β_2	.295	.298	.296	.074	.054	.313	.317	.311	.116	.040
β_3	.565	.559	.580	.046	.106	.481	.479	.480	.034	.122
β_4	.715	.718	.718	.167	.081	.717	.715	.716	.085	.091
γ_1	.196	.204	.204	.056	.042	.233	.229	.231	.017	.078
γ_2	.525	.536	.533	.063	.037	.478	.481	.478	.027	.044

$\mathrm{E}(x) = \sum_{1\le i<j\le n} x_{ij}$ is the number of edges capturing network density. $v(x) = e^{.2}\sum_{i=1}^{n-2}\{1 - (1-e^{-.2})^i\}D_i(x)$ is the alternating k-triangle statistic measuring friendship transitivity, and its ratio parameter is fixed to be .2 to facilitate the results comparison with MLE in Goodreau et al. (2008). $S(x, A)$ is a 3-dimensional vector of statistics capturing the similarities between friends on grade, race and gender, e.g., $\sum_{1\le i<j\le n} x_{ij}\mathbf{1}(grade_i = grade_j)$, where $\mathbf{1}(grade_i = grade_j) = 1$ if students i and j are in the same grade, and 0 otherwise. The prior we use is $N(0, 100I_5)$. We found it more efficient to work with edge list (a 1461×2 matrix whose rows recode dyads with edges) rather than a 1461×1461 matrix in programming. Again, we do a burn-in of 2000 and thinning of 10 for all results below.

To compare the performance of random sampler, TNT and OTNT, we carried out approximate sampling with each of them, respectively, starting with the same initial parameter values. Again we used $\kappa_n = \lceil d\log(1+n)\rceil$ with $d = 10$. For OTNT, we let $w_1 = .5, w_2 = w_3 = .25$. Figure 2 shows that in estimating alternating k-triangle parameter θ_2, OTNT produces much better mixing. The posterior sample of θ_2 is all over the place in random sampler, ranging from -10 to 30. While improved greatly comparing to random sampler, the mixing of TNT is still bad due to the lack of variation in the alternating k-triangle statistic value. On the contrary, OTNT mixes fast with a maximum κ_n as small as 93.

Table 4 gives the posterior means and 95% credible sets of the parameters. To save space, we only report results of the homogeneous instrument kernel, as both kernels yield very sim-

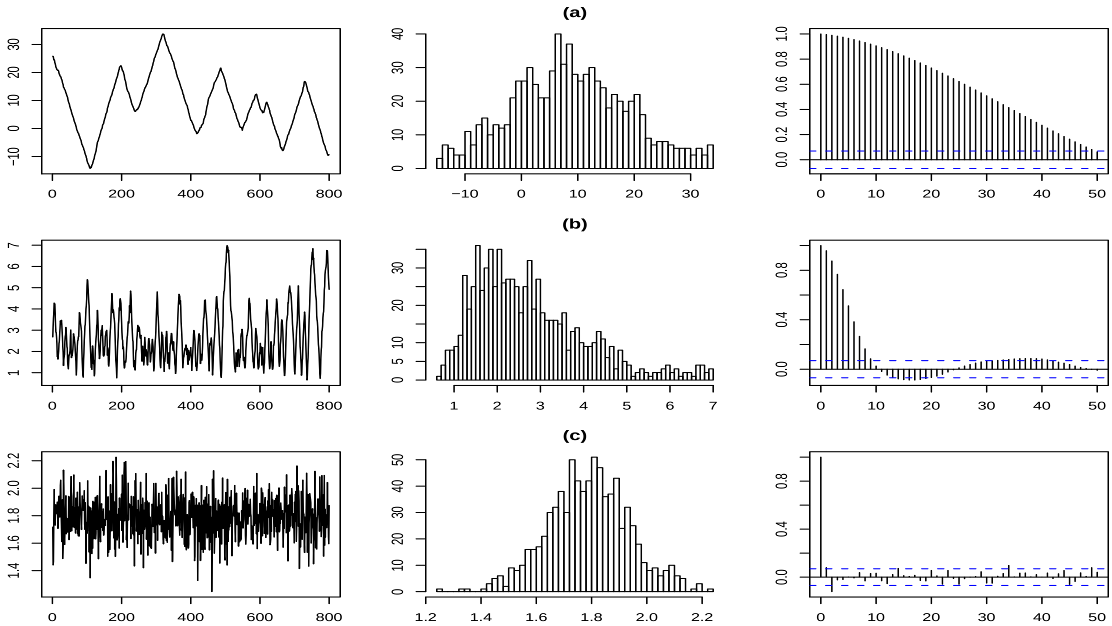

Figure 2: Social network example: trace plots, histograms and auto-correlation plots for θ_2 using different M-H moves (used in Algorithm 2.1 with the homogeneous instrument kernel). (a)-(c) are random sampler, TNT and OTNT, respectively.

ilar results. Approximate sampling with OTNT gives posterior means consistent with MLE in Goodreau et al. (2008). The Random sampler gives results that are completely off for all parameters, whereas the posterior mean of θ_2 for TNT is far from MLE as well. We believe that with much larger κ_n, TNT will eventually give a similar estimate, but at the expense of more computing power. We also present the posterior distributions for network density θ_1 and grade homophily β_1 generated with OTNT in figure 3. Again, the mixing is good, and posterior distributions are roughly normal with MLE close to posterior modes.

Appendix A. Proof of Theorem 2.1

Proof. Firstly, we will show that both $\bar{T}_\kappa$ and $\bar{T}_\star$ satisfy a uniform minorization condition. This implies the existence of a unique invariant distribution, and the kernel converges to the invariant distribution uniformly at a geometric rate (see e.g. Meyn and Tweedie (2003) Theorem 16.02).

We start with $\bar{T}_\star$ which is easier. Under the assumptions of the theorem and the compactness of Θ, we can easily find a probability density λ_Q on Θ, s.t. $\epsilon_Q \stackrel{\text{def}}{=} \inf_{\theta,\theta'\in\Theta} \frac{Q(\theta,\theta')}{\lambda_Q(\theta')} > 0$. Also, since

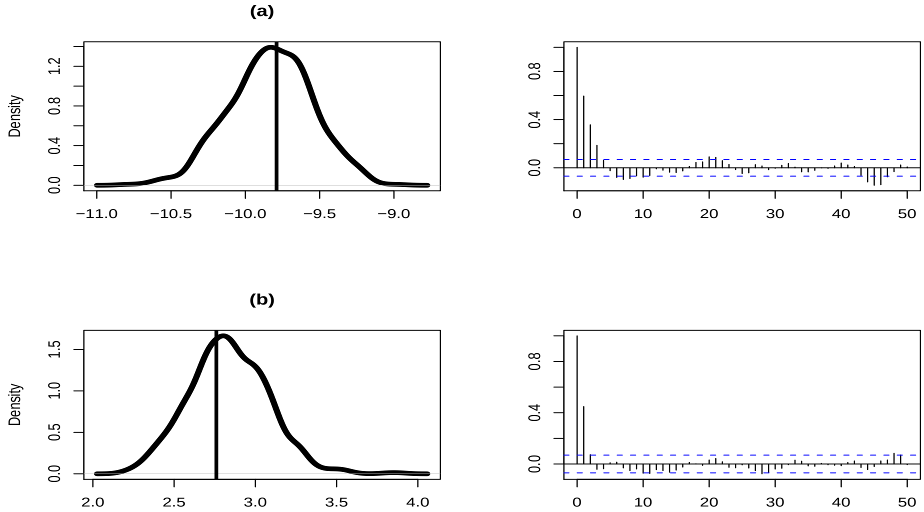

Figure 3: Social network example: density and autocorrelation plots of parameters corresponding to (a) network density θ_1 and (b) grade homophily β_1. Results are produced by algorithm 2.1 with homogeneous instrument kernel and OTNT. The vertical line indicates MLE in Goodreau et al. (2008).

Table 4: Social network example: the posterior means of Algorithm 2.1 with the homogeneous instrument kernel for different M-H moves compared with MLE in Goodreau et al. (2008).

Parameter	MLE	OTNT (95% Credible Set)	TNT	Random Sampler
Density (θ_1)	-9.79	-9.83 (-10.4, -9.28)	-9.88	-127
Alternating k-triangle (θ_2)	1.82	1.79 (1.50, 2.09)	2.75	8.65
Homophily grade effect (β_1)	2.76	2.83 (2.38, 3.29)	2.86	77.6
Homophily race effect (β_2)	.918	.926 (.421, 1.43)	.934	11.0
Homophily gender effect (β_3)	.766	.780 (.299, 1.25)	.806	6.94

Θ is compact and $\mathcal{X}$ is finite, there exists $C_\alpha \stackrel{\text{def}}{=} \inf_{\theta,\theta'\in\Theta} \inf_{y_\kappa\in\mathcal{X}} \alpha(\theta,\theta',y_\kappa) > 0$. Therefore

$$\bar{T}_\star\Big((\theta,x), A\times B\Big) \geq C_\alpha \epsilon_Q \int_A \lambda_Q(d\theta') \int_B p_{\theta'}(dx).$$

The argument is the same for $\bar{T}_\kappa$ but we need A1. We have

$$\bar{T}_\kappa\Big((\theta,x), A\times B\Big) \geq C_\alpha \epsilon_Q \int_{A\times B} \lambda_Q(d\theta') P_{\kappa,\theta,\theta'}(x, dy_\kappa).$$

Under A1,

$$\bar{T}_\kappa^{n_\kappa}\Big((\theta,x), A\times B\Big) \geq \left(C_\alpha \epsilon_Q |\mathcal{X}|\right)^{n_\kappa} \epsilon_\kappa \lambda_Q(A) \frac{|B|}{|\mathcal{X}|},$$

where $|B|$ denotes the number of elements of the set B. By Theorem 16.02 of (Meyn and Tweedie, 2003), $\bar{T}_\kappa$ and $\bar{T}_\star$ admit invariant distributions called $\bar{\pi}_\kappa$ and $\bar{\pi}_\star$ respectively, and for all $(\theta,x)\in\Theta\times\mathcal{X}$ and $n\geq 0$,

$$\left\|\bar{T}_\kappa^n\Big((\theta,x),\cdot\Big) - \bar{\pi}_\kappa(\cdot)\right\|_{\text{TV}} \leq (1-\bar{\epsilon}_\kappa)^{\lfloor n/n_\kappa \rfloor}, \tag{A.1}$$

$$\left\|\bar{T}_\star^n\left((\theta,x),\cdot\right) - \bar{\pi}_\star(\cdot)\right\|_{\text{TV}} \leq (1-\bar{\epsilon}_\star)^n, \tag{A.2}$$

where $\bar{\epsilon}_\kappa \stackrel{\text{def}}{=} \left(C_\alpha \epsilon_Q |\mathcal{X}|\right)^{n_\kappa} \epsilon_\kappa$, $\bar{\epsilon}_\star = C_\alpha \epsilon_Q$. We will use (A.1) repeatedly in Appendix B.

To prove (9), we first bound $\|\bar{T}_\kappa - \bar{T}_\star\|_{\text{TV}}$.

$$\begin{aligned}
& \left\|\bar{T}_\kappa\Big((\theta,x),\cdot\Big) - \bar{T}_\star\Big((\theta,x),\cdot\Big)\right\|_{\text{TV}} \\
\leq\ & \sup_{|h|\leq 1} \int_\Theta Q(\theta,d\theta') \left|\int_{\mathcal{X}} \left[P_{\kappa,\theta,\theta'}(x,dy_\kappa) - p_{\theta'}(dy_\kappa)\right] \alpha(\theta,\theta',y_\kappa) h(\theta',y_\kappa)\right| \qquad \text{(A.3)} \\
\leq\ & M_\alpha \int_\Theta Q(\theta,\theta') B_\kappa(\theta,\theta') d\theta', \qquad \text{(A.4)}
\end{aligned}$$

where $M_\alpha \stackrel{\text{def}}{=} \sup_{\theta,\theta'\in\Theta} \sup_{y_\kappa\in\mathcal{X}} \alpha(\theta,\theta',y_\kappa)$. Finally,

$$\begin{aligned}
\|\bar{\pi}_\kappa - \bar{\pi}_\star\|_{\text{TV}} &= \frac{1}{2}\sup_{|h|\leq 1} |\bar{\pi}_\kappa(\bar{T}_\star^n - \bar{\pi}_\star)h + \sum_{j=1}^n \bar{\pi}_\kappa(\bar{T}_\kappa - \bar{T}_\star)(\bar{T}_\star^{j-1} - \bar{\pi}_\star)h|, \\
\leq\ & \sum_{j\geq 0} (1-\bar{\epsilon}_\star)^j \sup_{\theta\in\Theta, x\in\mathcal{X}} \|\bar{T}_\kappa - \bar{T}_\star\|_{\text{TV}} \qquad \text{by letting } n\to\infty \text{ and } (A.2), \\
\leq\ & C \sup_{\theta\in\Theta} \int_\Theta Q(\theta,\theta') B_\kappa(\theta,\theta') d\theta', \qquad \text{by } (A.4), \text{ where } C = \frac{M_\alpha}{1-\bar{\epsilon}_\star}.
\end{aligned}$$

□

Appendix B. Proof of Theorem 2.2

Appendix B.1. Proof of Theorem 2.2 (1)

Proof. For $\kappa \geq 1$, we define $D_\kappa \overset{\text{def}}{=} C \sup_{\theta\in\Theta} \int_\Theta Q(\theta,\theta') B_\kappa(\theta,\theta') d\theta'$. We recall from the proof of Theorem 2.1 the quantity $\bar{\epsilon}_\kappa \overset{\text{def}}{=} (C_\alpha \epsilon_Q |\mathcal{X}|)^{n_\kappa} \epsilon_\kappa$. By assumption, $\rho \overset{\text{def}}{=} \sup_{\kappa\geq 1} 1 - \bar{\epsilon}_\kappa < 1$. Let $n_0 = \sup_{\kappa\geq 1} n_\kappa < \infty$.

For any $1 \leq j \leq n$ and measurable function $|h(\theta, x)| \leq 1$, we have

$$\left|\mathbb{E}\Big(h(\theta_n, X_n)\Big) - \bar{\pi}_\star(h)\right| = \left|\mathbb{E}\mathbb{E}\Big(h(\theta_n, X_n)|\theta_{j-1}, X_{j-1}\Big) - \bar{\pi}_\star(h)\right|,$$

so we first work with the conditional distribution of $\mathbb{E}\left(h(\theta_n, X_n)|(\theta_{j-1}, X_{j-1})\right)$.

$$\begin{aligned}
&\sup_{|h|\leq 1} \frac{1}{2} \left|\mathbb{E}\Big(h(\theta_n, X_n)|\theta_{j-1}, X_{j-1}\Big) - \bar{\pi}_\star(h)\right| \\
\leq\ & \|\bar{T}_{\kappa_j} \dots \bar{T}_{\kappa_n} - \bar{T}_{\kappa_n}^{n-j+1}\|_{\text{TV}} + \|\bar{T}_{\kappa_n}^{n-j+1} - \bar{\pi}_{\kappa_n}\|_{\text{TV}} + \|\bar{\pi}_{\kappa_n} - \bar{\pi}_\star\|_{\text{TV}}.
\end{aligned}$$

By (A.1) and (9), the last two terms are bounded by $\rho^{\lfloor \frac{n-j+1}{n_0} \rfloor} + D_{\kappa_n}$.

According to the following decomposition

$$(\bar{T}_{\kappa_j} \dots \bar{T}_{\kappa_n} - \bar{T}_{\kappa_n}^{n-j+1})h = \sum_{\ell=j-1}^{n-1} \bar{T}_{\kappa_j} \dots \bar{T}_{\kappa_\ell} (\bar{T}_{\kappa_{\ell+1}} - \bar{T}_{\kappa_n})(\bar{T}_{\kappa_n}^{n-\ell-1} - \bar{\pi}_{\kappa_n})h, \tag{B.1}$$

we bound the first term by $\sum_{\ell=j-1}^{n-1} \rho^{\lfloor \frac{n-j+1}{n_0} \rfloor} \|\bar{T}_{\kappa_{\ell+1}} - \bar{T}_{\kappa_n}\|_{\text{TV}}$.

Applying (A.4), we get $\|\bar{T}_{\kappa_{\ell+1}} - \bar{\pi}_\star\|_{\text{TV}} \leq D_{\kappa_{\ell+1}}$ and $\|\bar{T}_{\kappa_n} - \bar{\pi}_\star\|_{\text{TV}} \leq D_{\kappa_n}$. Therefore $\|\bar{T}_{\kappa_{\ell+1}} - \bar{T}_{\kappa_n}\|_{\text{TV}} \leq D_{\kappa_{\ell+1}} + D_{\kappa_n}$. Finally, we have

$$\sup_{|h|\leq 1} \frac{1}{2} \left|\mathbb{E}\left(h(\theta_n, X_n)\right) - \pi_\star(h)\right| \leq \sum_{\ell=j-1}^{n-1} \rho^{\lfloor \frac{n-j+1}{n_0} \rfloor} (D_{\kappa_{\ell+1}} + D_{\kappa_n}) + \rho^{\lfloor \frac{n-j+1}{n_0} \rfloor} + D_{\kappa_n}.$$

Let $j = \lceil n/2 \rceil$, under A2, this bound will go to 0 as $n \to \infty$, which completes the proof. □

Appendix B.2. Some preliminary results

We start with some general properties of Markov kernels, and prove Theorem 2.2 (2) afterwards. Let T_1, T_2 be two Markov kernels on a measurable general state space $(\mathbb{T}, \mathcal{B}(\mathbb{T}))$. Suppose $T_i, i = 1, 2$ satisfies a uniform minorization condition: there exist $\epsilon_i > 0$, probability measure λ_i, and a positive integer n_i, s.t. $T_i^n(w, B) \geq \epsilon_i^{\lfloor n/n_i \rfloor} \lambda_i(B), \forall B \in \mathcal{B}(\mathbb{T}), w \in \mathbb{T}$. Denote the corresponding invariant distribution by π_i (a general distribution on $\mathbb{T}$).

Proposition Appendix B.1. *We have*

a.

$$\|\pi_1 - \pi_2\|_{\mathrm{TV}} \le \frac{n_2}{\epsilon_2}\|T_1 - T_2\|_{\mathrm{TV}}. \tag{B.2}$$

b. *Let* $M_{h_i} \stackrel{\text{def}}{=} \sup_{w\in\mathbb{T}} |h_i(w)| < \infty$ *and* $\pi_i(h_i) = 0$, $i = 1, 2$ *and define*

$$g_i = \sum_{k\ge 0} T_i^k h_i.$$

Then $\sup_{w\in\mathbb{T}} |g_i| \le M_{h_i} n_i/\epsilon_i$ *and*

$$\sup_{w\in\mathbb{T}} |T_1 g_1 - T_2 g_2| \le K \left(\sup_{w\in\mathbb{T}} |h_1 - h_2| + \|T_1 - T_2\|_{\mathrm{TV}}\right), \tag{B.3}$$

where $K = \max\left(1 + \frac{n_1}{\epsilon_1}, \frac{M_{h_2} n_1 n_2}{\epsilon_1\epsilon_2}(1 + \frac{n_2}{\epsilon_2})\right)$.

Appendix B.3. Proof of Theorem 2.2 (2)

Proof.

$$n^{-1}\sum_{i=1}^n h(\theta_i, X_i) - \bar\pi_\star(h) = n^{-1}\sum_{i=1}^n h(\theta_i, X_i) - \bar\pi_{\kappa_i}(h) + n^{-1}\sum_{i=1}^n \bar\pi_{\kappa_i}(h) - \bar\pi_\star(h)$$

So we will prove both parts on the right hand side converge to 0 a.s., respectively.

First, we show $\lim_{n\to\infty} n^{-1}\sum_{i=1}^n h(\theta_i, X_i) - \bar\pi_{\kappa_i}(h) = 0$, a.s. Write $\bar h_i = h - \bar\pi_{\kappa_i}(h)$ and for any $\theta \in \Theta, x \in \mathcal{X}$, let $g_i(\theta, x) \stackrel{\text{def}}{=} \sum_{k\ge 0} \bar T_{\kappa_i}^k \bar h_i(\theta, x)$. Since g_i satisfies $g_i - \bar T_{\kappa_i} g_i = \bar h_i$, we have

$$\begin{aligned}
\sum_{i=1}^n \bar h_i(\theta_i, X_i) &= \sum_{i=1}^n g_i(\theta_i, X_i) - \bar T_{\kappa_i} g_i(\theta_{i-1}, X_{i-1}) + \left(\bar T_{\kappa_1} g_1(\theta_0, X_0) - \bar T_{\kappa_n} g_n(\theta_n, x_n)\right) \\
&\quad + \sum_{i=2}^n \bar T_{\kappa_i} g_i(\theta_{i-1}, X_{i-1}) - \bar T_{\kappa_{i-1}} g_{i-1}(\theta_{i-1}, X_{i-1}) \\
&= \sum_{i=1}^n g_i(\theta_i, X_i) - \bar T_{\kappa_i} g_i(\theta_{i-1}, X_{i-1}) + R_{1,n}.
\end{aligned}$$

By Proposition Appendix B.1, we have $\sup_{i\ge 1} |g_i| < \infty$. Combining (B.2), (B.3), and Kronecker's lemma (Hall and Heyde, 1980), it follows that $n^{-1}R_{1,n} \to 0$. Let $D_i = g_i(\theta_i, X_i) - \bar T_{\kappa_i} g_i(\theta_{i-1}, X_{i-1})$. It is easy to see that each D_i is a martingale difference. By Proposition Appendix B.1 and Minkowski's inequality, we have $\sum_{i=1}^\infty i^{-2}\mathbb{E}D_i^2 < \infty$. Therefore, $n^{-1}\sum_{i=1}^n D_i \to 0$, a.s. (Chow, 1960). This proves that $n^{-1}\sum_{i=1}^n h(\theta_i, X_i) - \bar\pi_{\kappa_i} h \to 0$, a.s.

Next we show $\lim_{n\to\infty} n^{-1}\sum_{i=1}^{n} \bar{\pi}_{\kappa_i}(h) - \bar{\pi}_\star h = 0$. By Theorem 2.1, we have

$$|\bar{\pi}_{\kappa_i} h - \bar{\pi}_\star h| \le 2M_h D_{\kappa_i}.$$

Kronecker's lemma concludes the proof of the strong law of large numbers.

For central limit theorem, again we have the martingale approximation

$$\sum_{i=1}^{n} \Big(h(\theta_i, X_i) - \bar{\pi}_\star h \Big) = \sum_{i=1}^{n} g_i(\theta_i, x_i) - \bar{T}_{\kappa_i} g_i(\theta_{i-1}, x_{i-1}) + R_{2,n},$$

where $R_{2,n} = R_{1,n} + \sum_{i=1}^{n}(\bar{\pi}_{\kappa_i} h - \bar{\pi}_\star h)$. By (13) and Kronecker's lemma, we have $n^{-1/2}R_{2,n}$ converges to 0. The term $\sum_{i=1}^{n} D_i$ is a triangular martingale array. Since $\sup_{i\ge 1}\sup_{\theta,x} D_i^2 < \infty$, the conditional Lindeberg condition holds.

To show $n^{-1}\sum_{i=1}^{n} \bar{T}_{\kappa_i} D_i^2 \to \sigma^2(h)$ a.s., we use a similar logic as above. But instead of $h(\theta, x)$, we prove a strong law of large numbers for function $q_i(\theta, x) = \bar{T}_{\kappa_i} D_i^2 = \bar{T}_{\kappa_i} g_i^2(\theta, x) - \big(\bar{T}_{\kappa_i} g_i(\theta, x)\big)^2$. Since $\sup_{i\ge 1} M_{h_i} < \infty$, we can show

$$\lim_{n\to\infty} n^{-1} \sum_{i=1}^{n} q_i(\theta_{i-1}, X_{i-1}) - \bar{\pi}_i q_i = 0, \quad \text{a.s.}$$

Moreover, it is not hard to show that for any $k \ge 1$,

$$\lim_{n\to\infty} n^{-1} \sum_{i=1}^{n} \bar{\pi}_{\kappa_i}(h \bar{T}_{\kappa_i}^k h) - \bar{\pi}_\star(h \bar{T}_\star^k h) = 0, \quad \text{a.s.}$$

Therefore, we have

$$\lim_{n\to\infty} n^{-1} \sum_{i=1}^{n} \bar{T}_{\kappa_i} g_i^2(\theta_{i-1}, X_{i-1}) - \big(\bar{T}_{\kappa_i} g_i(\theta_{i-1}, X_{i-1})\big)^2 = \sigma^2(h) \quad \text{a.s.},$$

where $\sigma^2(h)$ is defined in Theorem 2.2. By Corollary 3.1 in Hall and Heyde (1980) we deduce that

$$\frac{1}{\sqrt{n}} \sum_{i=1}^{n} \Big(h(\theta_i, X_i) - \bar{\pi}_\star h \Big) \xrightarrow{\mathcal{D}} N\left(0, \sigma^2(h)\right).$$

□

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